



Mastering Loops

The Logic of Repetition and Efficiency

Why Do We Need Loops?

WITHOUT LOOPS

Manual Repetition

Grade Student #1

Grade Student #2

Grade Student #3

30 Blocks of Redundant Code

WITH LOOPS

Automated Iteration

FOR each student:

grade(student)

1 Block Executed 30 Times

Selecting the Right Structure



for loop

"I know exactly" how many times to repeat.

Input: A collection/list

- Processing each row
- Iterating through range(10)



while loop

"I know when to stop" based on a condition.

Input: A Boolean test

TRUE → Repeat

FALSE → Exit

Anatomy of a Loop

① INITIALIZATION

```
count = 0
```



② CONDITION CHECK

```
while count < 5:
```



③ BODY + ④ UPDATE

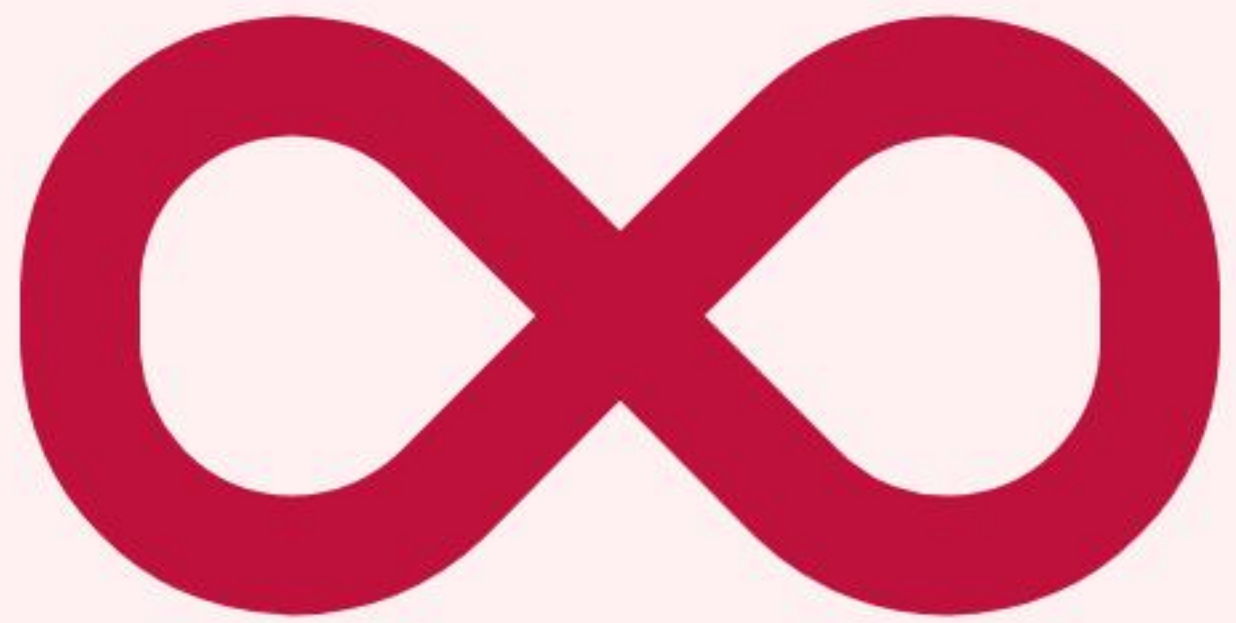
```
print(count) # The Work
```

```
count = count + 1 # The Progress
```



→ NO → EXIT

| The Infinite Loop Trap



```
count = 0
while count < 5:
    print(count)
    # UPDATE MISSING!
```

Logic Error:

Because count stays 0, it is *always* less than 5. The loop never breaks, leading to a program hang or crash.

Visualizing range()

RANGE(5)

Starts at 0, stops **before** 5.

0

1

2

3

4

RANGE(2, 10, 2)

Start: 2, Stop: 10, **Step: 2**

2

4

6

8

range(start, stop, step) is the most powerful tool for for-loop control.

Loop Pattern: Accumulation

Building a Total

Loops are frequently used to "accumulate" values into a single variable.

```
total = 0
for num in list:
    total = total + num
```

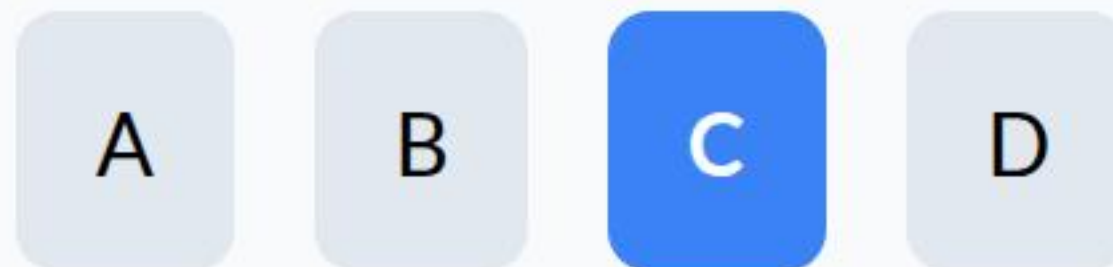
Visual Flow

[1] → +1 → [2] → +2 →
[4] → +3 → 7

Loop Pattern: Searching



Finding 'C'



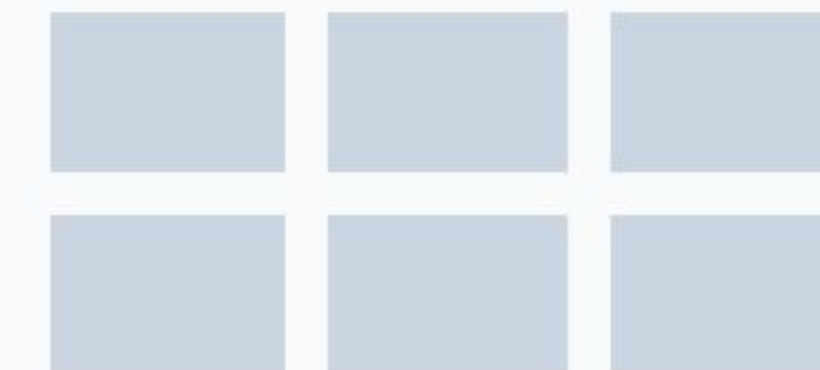
Logic: Match found? → STOP (break)



Nested Logic

Outer Loop: Rows

Inner Loop: Columns



Mistake 1: Off-By-One

OFF-BY-ONE LOGIC

Problem: Want to print 1 to 5.

✗ `range(5)` → 0, 1, 2, 3, 4

✗ `range(1, 5)` → 1, 2, 3, 4

✓ `range(1, 6)` → 1, 2, 3, 4, 5

Analogy: If you have a fence with 5 panels, you need 6 posts.



Mistake 2: List Shifts

MODIFYING DURING ITERATION

Before



After Removal



Warning: The list shifts! The loop might skip item [3] because its index changed while you were iterating.

Which Loop to Use?

The Decision Question:

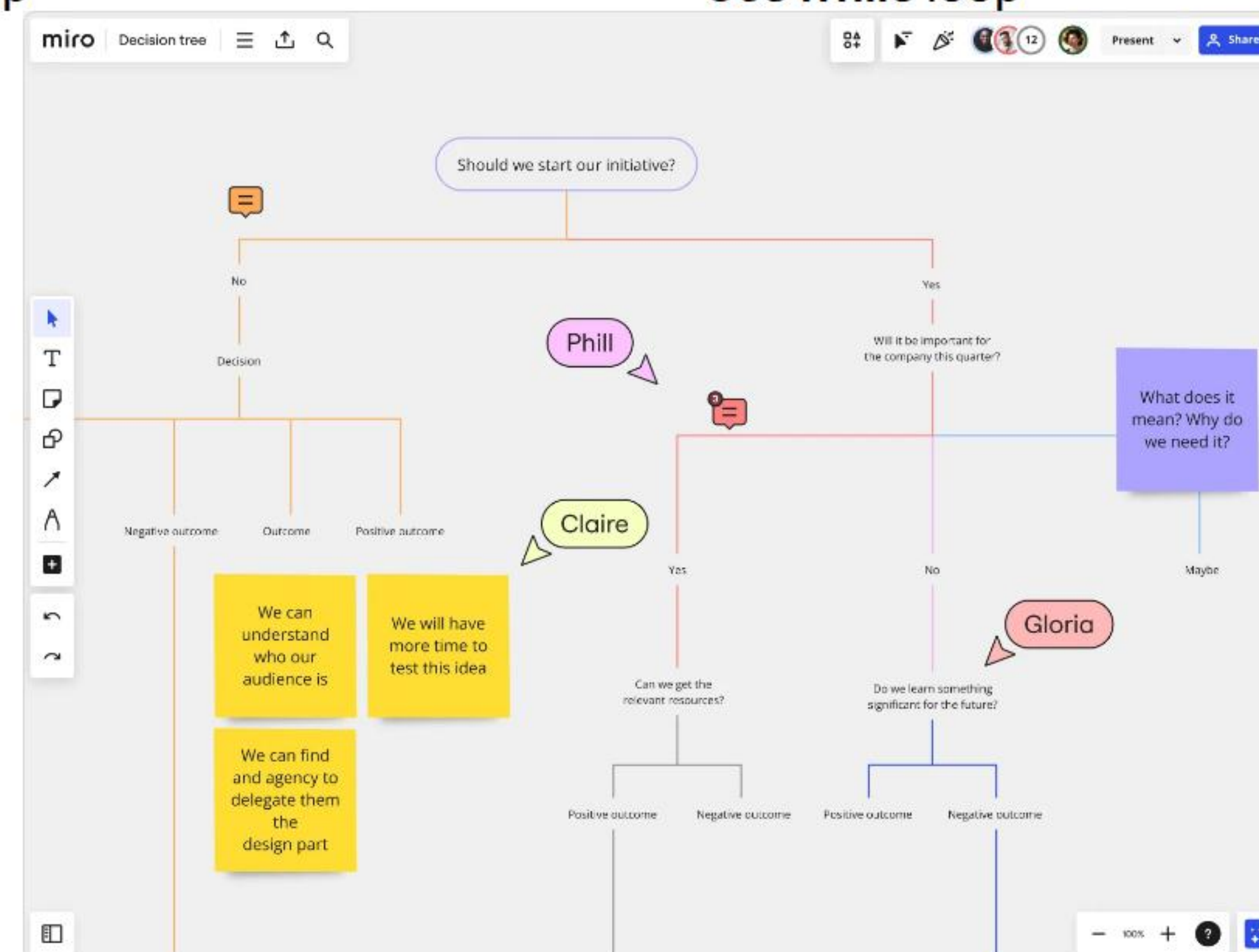
Do I know **BEFORE** starting how many times the code needs to run?

YES

Use **for** loop

NO

Use **while** loop



Execution Trace: "Hello!" 5x

Iteration	Variable (i)	Body Action	Result
Pass 1	0	print("Hello!")	Hello!
Pass 2	1	print("Hello!")	Hello!
Pass 3	2	print("Hello!")	Hello!
Pass 4	3	print("Hello!")	Hello!
Pass 5	4	print("Hello!")	Hello!

— Final Output: 5 Lines of Hello! —

Image Sources



https://www.softwareseni.com/wp-content/uploads/2025/09/CSW202509300852_ART005_header_image.png

Source: www.softwareseni.com



https://images.ctfassets.net/w6r2i5d8q73s/5SSgDmtdB3h5tUEzuOE49r/0c71f2b075114b02e36768b77caf297d/decision_treemaker_header_image_EN_standard_4_3.png?fm=webp&q=75

Source: miro.com